Natural Language Processing: Assignment 2

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1 Dataset & Setup

To classify news articles into one of four categories (world, sports, business, and sci/tech), we use the AG News dataset from Hugging Face [Zhang et al., 2015]. This dataset consists of news articles with associated labels, titles, and descriptions. Each class contains 30,000 training samples and 1,900 testing samples, resulting in a total of 120,000 training samples and 7,600 testing samples.

We used the official Hugging Face split (120,000 train and 7,600 test). The training set was split into a new training set (of size 108,000) and a development set (of size 12,000) using stratified sampling with a fixed random seed of 67 to ensure reproducibility. The final split sizes are: train 108,000, dev 12,000, and test 7,600.

Preprocessing. We combined the title and description of each news article into a single text input. We cleaned the text by HTML unescaping to convert HTML entities back to their original characters (like `#39;` to `'`) and removed any remaining HTML tags. We also unescaped LaTeXentities (\\ etc.), collapse backslashes and normalize whitespace. We converted the labels from 1-based to 0-based indexing (i.e., world: 0, sports: 1, business: 2, sci/tech: 3) for easier handling in the classification models.

Tokenization. We used Byte Pair Encoding (BPE) tokenization, which is a subword tokenization method that can effectively handle out-of-vocabulary words and reduce the vocabulary size while still capturing meaningful subword units. We trained a BPE tokenizer training set with a vocabulary size of 5,000 and a minimum frequency of 2 to ensure that only subwords that appear at least twice in the training data are included in the vocabulary. We convert token sequences to fixed-length inputs via truncation/padding to a maximum sequence length (64 vs. 128 vs. 256).

2 Models

In this assignment, we compare two neural models for news classification: a Convolutional Neural Network (CNN) and a Long Short-Term Memory network (LSTM). To make the comparison fair, both models use the same data splits, preprocessing pipeline, and evaluation setup.

Shared input representation. Both models receive the same preprocessed and tokenized text (see Section 1 for preprocessing and tokenization details). Sequences are padded or truncated to the chosen maximum length. Both models begin with a trainable embedding layer that maps token ids to dense vectors. Note that the CNN has inherent feature extraction through its convolutional layers, while the LSTM relies on the embeddings and recurrent connections for representation learning.

CNN model. The CNN is based on the text classification model of [Kim, 2014]. After the embedding layer, the model applies several 1-dimensional convolution filters in parallel. These filters are used to detect informative local patterns in the text, such as short phrases or combinations of nearby words. We use multiple kernel sizes so the model can capture patterns of different lengths. After the convolution step, a ReLU activation function is applied, followed by max-pooling to keep the strongest signal from each filter. The resulting features are then concatenated, followed by dropout for regularisation, and finally passed to a linear layer that predicts one of the four classes.

LSTM model. The LSTM model is based on the Long Short-Term Memory architecture introduced by [Hochreiter and Schmidhuber, 1997]. It reads the input sequence token by token, which makes it suitable for modelling word order and longer dependencies in the text. We use a stacked 2-layer LSTM. To avoid learning from padding tokens, we handle variable-length sequences by skipping over padded positions during the recurrent computation, so the model only processes the real tokens in each example. For classification, we use the final hidden state of the top LSTM layer as the representation of the whole article. In the bidirectional setting, the model combines information from both directions before making the final prediction. As in the CNN, dropout is applied before the final linear layer.

Hyperparameters. Table 1 shows the hyperparameter search space for both models. For each model, the best setting was selected based on macro-F1 on the development set.

Parameter	**CNN**	**LSTM**
Embedding dim d	{128, 512}	{128, 256}
Filters F	{100, 300}	—
Kernel sizes K	{[3,5], [3,5,7]}	—
Hidden dim H	—	{128, 256, 512}
Bidirectional	—	{False, True}
Dropout	{0.3, 0.5}	0.5
Learning rate	{1e-3, 1e-4}	{1e-3, 1e-4}
Weight decay	{1e-5, 1e-3}	{0, 1e-5}

Table 1: Hyperparameter search spaces for the CNN and LSTM models.

Shared training settings. Both models were trained with the Adam optimiser. We used gradient clipping with a maximum norm of 5.0 and early stopping with a patience of 3 epochs based on validation loss. The batch size was set to 512, and the final training runs lasted for at most 15 epochs. During hyperparameter tuning, each configuration was trained for 3 epochs. All experiments were run on the Hábrók high-performance computing cluster at the University of Groningen, using one GPU per job and 8 GB of RAM.

Ablation: maximum sequence length. As our ablation study, we vary the maximum sequence length and compare settings of 64, 128, and 256 tokens. All other parts of the setup remain the same. This allows us to study how much context the models need and whether giving them longer input sequences improves performance enough to justify the extra computation.

3 Experimental Protocol

For each model, we first performed hyperparameter tuning on the training and development sets. The CNN and LSTM were tuned separately using grid search over the parameter ranges shown in Table 1. During

tuning, each configuration was trained for 3 epochs, and the best configuration was selected based on macro-F1 on the development set. Early stopping with a patience of 3 epochs was enabled during this tuning stage, although in practice the short tuning runs meant that most configurations finished quickly.

After selecting the best hyperparameters for a given model and sequence length, we trained that model again using the full training portion of the data and evaluated it on the unchanged test set. The final training runs used the Adam optimiser, a batch size of 512, gradient clipping with a maximum norm of 5.0, and a maximum of 15 epochs. Unlike the tuning stage, the final training procedure did not apply early stopping in code; instead, models were trained for the full configured number of epochs and then evaluated on the development and test sets.

We report accuracy and macro-F1 as our main evaluation metrics, with macro-F1 used as the primary measure for model selection because it gives equal weight to all four classes. In addition, we generated classification reports, confusion matrices, learning curves, and files with misclassified test examples for later error analysis. The test set was only used for final evaluation after model selection.

As our ablation study, we varied the maximum input sequence length while keeping the rest of the setup fixed. We ran each model with maximum sequence lengths of 64, 128, and 256 tokens, resulting in six main experiment settings: CNN-64, CNN-128, CNN-256, LSTM-64, LSTM-128, and LSTM-256. These experiments were executed on the Hábrók cluster as a job array, with one GPU assigned to each run. This design allows us to compare the two architectures directly and also study how much performance depends on the amount of input context available to the model.

4 Results

4.1 Best hyperparameters

Table 2 summarises the best hyperparameters found during the tuning phase for each model and sequence length. The CNN consistently preferred a large embedding dimension (512), a high number of filters (300), kernel sizes [3, 5, 7], and a learning rate of 0.001. The LSTM favoured lower embedding dimensions (128–256) and hidden dimensions (128–256), also with a learning rate of 0.001. Interestingly, both architectures benefited from very low weight decay (1e-5 or 0.0).

Run	**Embed**	**Filters**	**Hidden**	**BiLSTM**	**LR**	**Dropout**	**WD**
CNN-64	512	300	—	—	0.001	0.3	1e-5
CNN-128	512	300	—	—	0.001	0.5	1e-5
CNN-256	512	300	—	—	0.001	0.5	1e-5
LSTM-64	256	—	128	Yes	0.001	0.5	1e-5
LSTM-128	128	—	256	Yes	0.001	0.5	0.0
LSTM-256	256	—	128	Yes	0.001	0.5	0.0

Table 2: Best hyperparameter configurations for each run. Embed: embedding dimension, WD: weight decay.

4.2 Test set performance

The test results are shown in Table 3. The CNN model achieved the highest performance at all three sequence lengths. At the maximum sequence length of 256 tokens, the CNN reached an accuracy of 90.79% and a macro-F1 of 90.78%. The LSTM performed comparably well, with its best result also at sequence length 256 (accuracy 90.71%, macro-F1 90.72%).

Both models improved as the maximum sequence length increased from 64 to 256 tokens. For the CNN, macro-F1 increased from 90.59% to 90.78%. For the LSTM, macro-F1 increased from 89.58% to 90.72%. The difference between the two models was largest at sequence length 64 and smallest at sequence length 256.

Model	Seq Len	Dev F1	Test Acc	Macro F1
CNN-64	64	0.9025	0.9061	0.9059
CNN-128	128	0.9061	0.9049	0.9047
CNN-256	256	**0.9084**	**0.9079**	**0.9078**
LSTM-64	64	0.8986	0.8962	0.8958
LSTM-128	128	0.9026	0.8978	0.8980
LSTM-256	256	0.9082	0.9071	0.9072

Table 3: Test set accuracy and macro-F1 scores. Dev F1 is the best development macro-F1 found during tuning.

4.3 Ablation study: effect of maximum sequence length

Figure 1 shows how test accuracy and macro-F1 vary with sequence length.

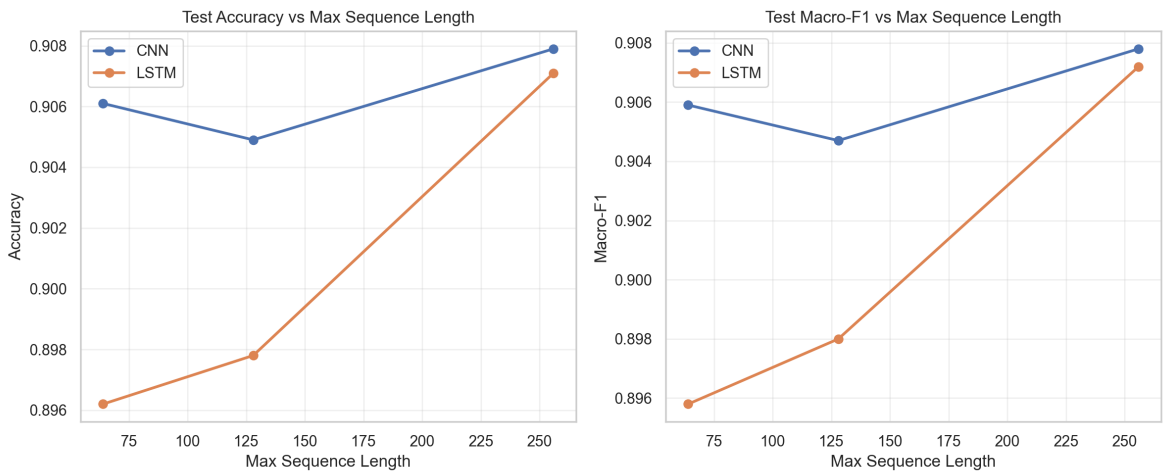

Figure 1: Test accuracy and macro-F1 as a function of maximum sequence length for CNN and LSTM.

The overall trend is clear: both CNN and LSTM improve as the maximum sequence length increases from 64 to 256 tokens. The CNN is slightly better than the LSTM at all three sequence lengths. At seq64, the gap is about 1 percentage point in macro-F1, while at seq256 the gap becomes very small (90.78% vs. 90.72%).

4.4 Class-wise analysis

Figure 2 presents the per-class macro-F1 scores.

The World and Sports classes are generally easier to classify, with F1 scores above 92% in most configurations. The Sci/Tech class is slightly harder, with F1 scores typically around 86–88%. The most challenging class is Business, where F1 scores range from 85% to 87%, depending on the model and sequence length.

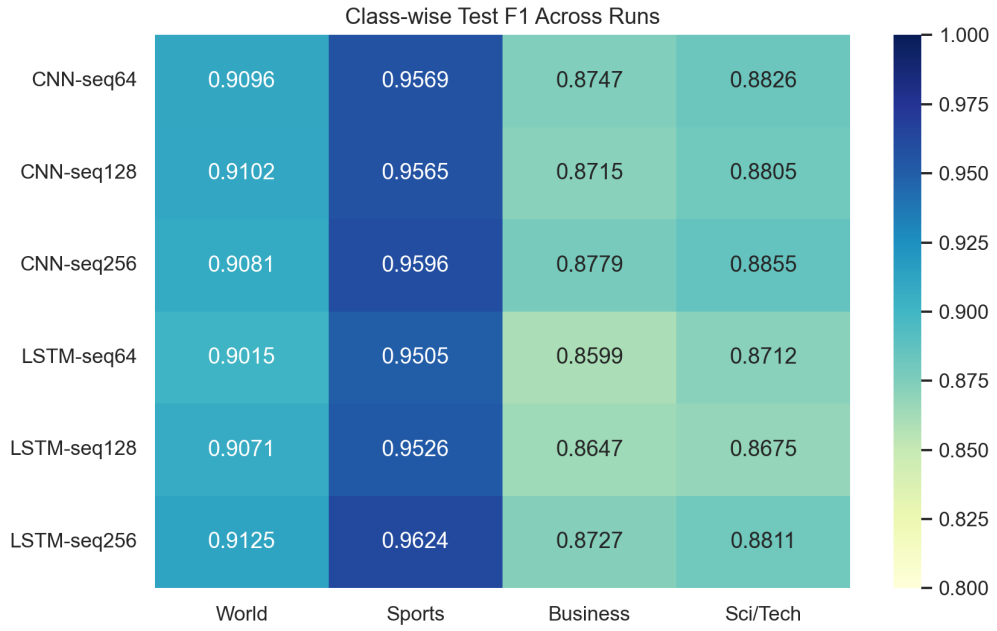

Figure 2: Heatmap of per-class test F1 scores across all runs. World and Sports are easier to classify; Business is the most challenging.

4.5 Learning curves

Figure 3 displays the learning curves for the final training runs (with best hyperparameters on the full training set).

Both models show rapid convergence, with validation loss decreasing sharply in the first few epochs and then stabilizing. The CNN typically reaches its best validation performance around epoch 8–10, while the LSTM reaches it slightly later (epochs 10–12). In the later epochs, the LSTM curves show larger validation-loss fluctuations than the CNN curves.

4.6 Summary

The CNN model achieves the best overall result, but the margin over the LSTM is small, especially at the longest sequence length. Both models benefit from longer input sequences. The best overall configuration is CNN-seq256 with a test macro-F1 of 90.78%.

5 Error Analysis

To better understand the remaining errors, we examined the first 20 misclassified test examples saved for each of the six runs. Rather than looking at isolated mistakes, we focused on examples that reappeared across multiple models and sequence lengths. This makes it possible to identify systematic weaknesses instead of noise from a single run. Full listings of all misclassified examples are provided in Appendix A.

The clearest pattern is confusion between *Sci/Tech* and *Business*. Across the saved error files, *Sci/Tech* articles misclassified as *Business* is the single most common error type, followed by the reverse. This is

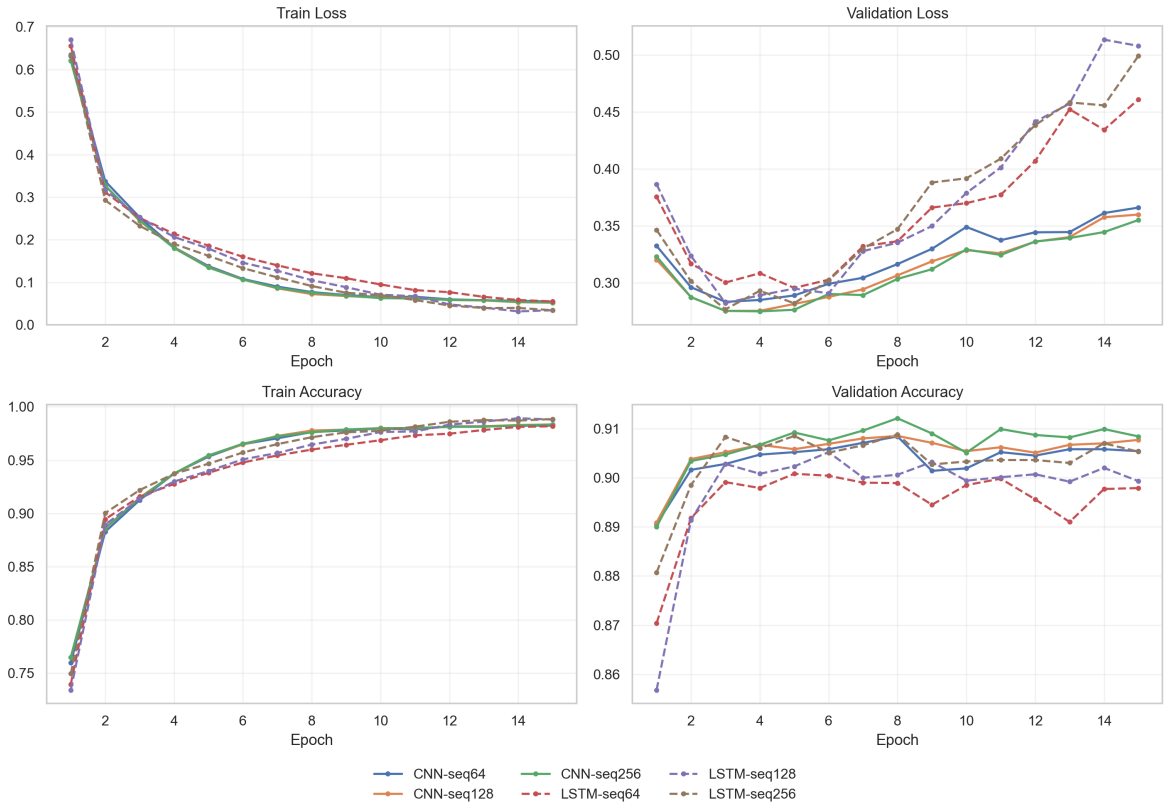

Figure 3: Training and validation loss/accuracy curves for all six runs. Both models show rapid early convergence and stable learning trajectories.

unsurprising: many technology articles in AG News employ business language such as *market*, *shares*, *growth*, or *workers*, and frequently reference company names. Sci/Tech articles that were misclassified as Business include "IBM to hire even more new workers" (App. A.1, Example 5), "Dell Exits Low-End China Consumer PC Market" (App. A.1, Example 8), and a Google IPO article (App. A.1, Example 6). Conversely, Business articles mislabeled as Sci/Tech include "Intel to delay product aimed for high-definition TVs" (App. A.1, Example 11), "Yahoo! Ups Ante for Small Businesses" (App. A.1, Example 15), and "Surviving Biotech's Downturns" (App. A.1, Example 13). These systematic errors reveal that the boundary between technological innovation and economic reporting is inherently ambiguous, especially in news data.

Another recurring pattern is confusion between *World* and *Business*. Articles about politics or international affairs are often misclassified as *Business* when they mention oil prices, trade, or financial consequences. Examples include "Oil prices bubble to record high" (App. A.1, Example 16), "Venezuela Prepares for Chavez Recall Vote" (App. A.2, Example 7), and "Google Lowers Its IPO Price Range" (App. A.1, Example 18). In these cases, the model seems to rely strongly on economic keywords, even when the broader topic is political or international. This suggests that topical overlap between global affairs and economics remains difficult for both architectures.

We also observe a smaller but consistent confusion between *World* and *Sports*. For instance, "Live: Olympics day four" (App. A.1, Example 10) is repeatedly predicted as *Sports* although it is labelled as *World*, and political or public-affairs articles such as "Mr Downer shoots his mouth off" (App. A.1, Example 14) are sometimes predicted as *Sports*. These mistakes suggest that short headlines without much context can be difficult to classify, especially when they contain named entities or event-style phrasing that does not clearly reveal the topic.

Some errors are caused by lexical ambiguity inside very short texts. The article "E-mail scam targets police chief" (App. A.2, Example 2) was misclassified as *World* in multiple runs, probably because words such as *police* and *chief* are also common in political or crime-related world news. Similarly, "Super ant colony hits Australia" (App. A.1, Example 3) was often predicted as *World*, even though it belongs to *Sci/Tech*. In these examples, the model appears to focus on surface cues such as country names or institutional words rather than the underlying scientific content.

Comparing the two models, the overall error types are very similar, but the CNN tends to make slightly fewer mistakes on the hardest category boundaries. The LSTM shares most of the same repeated errors, especially the *Sci/Tech–Business* confusions, which suggests that additional sequence modelling alone is not enough to solve these cases. Many of the problematic examples are difficult because they are genuinely mixed-topic articles rather than simple classification failures.

The saved errors also show that several examples are misclassified in almost every run. Articles about IBM, Yahoo!, Intel, oil prices, and Google IPOs appear repeatedly across both models and all sequence lengths, indicating that these cases reflect persistent ambiguities in the dataset rather than random noise from individual runs.

Overall, the error analysis suggests that the main challenge in this task is separating articles that sit near the boundary between two related topics. In particular, the distinction between *Sci/Tech* and *Business* remains the most difficult for both models. This is consistent with the class-wise results reported in Section 4, where Business and Sci/Tech were the lowest-performing categories.

6 Discussion

Our results suggest that this task depends a lot on short local patterns in the text. That helps explain why the CNN is already strong at sequence length 64: even with shorter inputs, it can pick up useful topic phrases and keywords. So the CNN advantage at short length is not only a score difference, but also a good match between the model and the kind of signal in AG News.

At the same time, the BiLSTM improves more as we increase sequence length, and at 256 tokens it almost catches the CNN (Table 3). A likely reason is that the extra context helps the LSTM in examples where early words are not enough to decide the class. This seems especially relevant for Business and Sci/Tech, where articles often contain mixed vocabulary. Still, even at 256 the LSTM does not clearly pass the CNN, which suggests that local cues remain very important for this dataset.

One result that we should explicitly discuss is the CNN-128 anomaly. In Table 3, CNN-128 is slightly worse than CNN-64 on test accuracy and macro-F1 (0.9049 vs. 0.9061 and 0.9047 vs. 0.9059). So the sequence-length ablation is not perfectly monotonic. We cannot fully prove the cause from one run, but the difference is very small (about 0.1 points) and could come from normal tuning/run variance together with a less favorable hyperparameter setting for that run.

The class-wise results support this broader picture. World and Sports are generally easier for both models, while Business and Sci/Tech remain harder. This tells us that the remaining errors are mostly about separating semantically close topics, not about basic model failure.

Overall, CNN-seq256 is still the best single configuration in our experiments, but the margin over LSTM-256 is small. However, although the CNN is better, it does not scale with sequence length as well as the LSTM, we think that if sequence length were increased further, the LSTM might eventually outperform the CNN. This is because the LSTM can potentially leverage longer-range dependencies and more global context, while the CNN relies more on local patterns that may saturate in usefulness after a certain point. In future work, it would be interesting to explore even longer sequence lengths and see if the LSTM can eventually surpass the CNN as it gets more context to work with.

7 Limitations

This study has a few limitations. First, we only evaluate on AG News, so our conclusions may not transfer to datasets with different writing styles or label structures. A second limitation is our choice of a 5,000-item BPE vocabulary. This is a reasonable size for a small study, but we did not check token coverage on the test set, so we cannot say how much information may have been lost because of vocabulary size. A larger vocabulary might have preserved more useful distinctions, especially for named entities and domain-specific terms.

Another important limitation is the tuning setup. Hyperparameter search was done with only 3 epochs per configuration, and we did not explicitly verify that this was long enough for strong and weak configurations to separate clearly. This means some settings may have looked worse simply because they learned more slowly, not because they were truly worse. The CNN-128 result also shows that small differences between runs should be interpreted carefully.

Regularisation is another place where our choices may have been too limited. In the best configurations, weight decay was either very small or zero, and the learning curves suggest that the LSTMs overfit slightly more than the CNNs. Because of this, it would have been useful to test higher weight decay values more aggressively, especially for the LSTM models.

Finally, we compare only CNN and LSTM models trained from scratch, so this is a comparison between two classical neural architectures rather than a comparison with the current strongest text-classification models. That does not affect the internal comparison in this report, but it does limit how far the absolute performance numbers can be taken.

References

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[Kim, 2014] Kim, Y. (2014). Convolutional neural networks for sentence classification. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1746–1751, Doha, Qatar. Association for Computational Linguistics.

[Zhang et al., 2015] Zhang, X., Zhao, J., and LeCun, Y. (2015). Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28 (NIPS 2015)*.

A Detailed Misclassification Examples

This appendix provides a comprehensive listing of all misclassified examples from the test set across all six experimental runs. Each example is labeled with its dataset index, original label, and predicted label for reference in the main report.

A.1 CNN (Max Sequence Length: 64)

Total misclassified examples: 20

Example 1 *(Index 0: Gold: Business, Predicted: World)*

> *Fears for T N pension after talks Unions representing workers at Turner Newall say they are 'disappointed' after talks with stricken parent firm Federal Mogul.*

Example 2 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

> *Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.*

Example 3 *(Index 13: Gold: Sci/Tech, Predicted: World)*

> *Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.*

Example 4 *(Index 19: Gold: Sci/Tech, Predicted: Business)*

> *Storage, servers bruise HP earnings update Earnings per share rise compared with a year ago, but company misses analysts' expectations by a long shot.*

Example 5 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

> *IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.*

Example 6 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

> *Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of public participation. But this isn't the case.*

Example 7 *(Index 24: Gold: Sci/Tech, Predicted: Business)*

Rivals Try to Turn Tables on Charles Schwab By MICHAEL LIEDTKE SAN FRANCISCO (AP) – With its low prices and iconoclastic attitude, discount stock broker Charles Schwab Corp. (SCH) represented an annoying stone in Wall Street's wing-tipped shoes for decades...

Example 8 *(Index 41: Gold: Business, Predicted: World)*

Retailers Vie for Back-To-School Buyers (Reuters) Reuters - Apparel retailers are hoping their back-to-school fashions will make the grade among style-conscious teens and young adults this fall, but it could be a tough sell, with students and parents keeping a tighter hold on their wallets.

Example 9 *(Index 55: Gold: Sci/Tech, Predicted: Sports)*

Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.

Example 10 *(Index 79: Gold: World, Predicted: Sports)*

Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.

Example 11 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

Example 12 *(Index 89: Gold: Sci/Tech, Predicted: Business)*

Consumers Would Pay In Phone Proposal A proposal backed by a coalition of telephone carriers would cut billions of dollars in fees owed by long-distance companies to regional phone giants but would allow the regional companies to make up some of the difference by raising monthly phone bills for millions of consumers. ¡FONT face="verdana,MS Sans Serif,arial,helvetica" size="-2" color="#666666"¿¡B¿-The Washington Post¡/B¿¡/FONT¿

Example 13 *(Index 103: Gold: Business, Predicted: Sci/Tech)*

Surviving Biotech's Downturns Charly Travers offers advice on withstanding the volatility of the biotech sector.

Example 14 *(Index 104: Gold: World, Predicted: Sports)*

Mr Downer shoots his mouth off Just what Alexander Downer was thinking when he declared on radio last Friday that quot;they could fire a missile from North Korea to Sydney quot; is unclear. The provocative remark, just days before his arrival yesterday on his second visit to the North Korean ...

Example 15 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space – which means that small businesses looking to move online get a sweeter deal through ...

Example 16 *(Index 120: Gold: World, Predicted: Business)*

Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 17 *(Index 139: Gold: World, Predicted: Business)*

Sharon orders 1,000 homes in West Bank Israel announced plans for 1,000 houses in the West Bank yesterday, accelerating the expansion of the settlements.

Example 18 *(Index 154: Gold: World, Predicted: Business)*

Google Lowers Its IPO Price Range SAN JOSE, Calif. - In a sign that Google Inc.'s initial public offering isn't as popular as expected, the company lowered its estimated price range to between $85 and $95 per share, down from the earlier prediction of $108 and $135 per share...

Example 19 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 20 *(Index 172: Gold: Business, Predicted: Sci/Tech)*

Ross Stores Profit Plummets 40 Percent (AP) AP - Discount retailer Ross Stores Inc. Wednesday said its profit fell about 40 percent in the latest quarter due to problems with a new computer system that limited the company's ability to respond to changes in customer demand.

A.2 CNN (Max Sequence Length: 128)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.

Example 2 *(Index 8: Gold: Sci/Tech, Predicted: World)*

E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.

Example 3 *(Index 13: Gold: Sci/Tech, Predicted: World)*

Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.

Example 4 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.

Example 5 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of public participation. But this isn't the case.

Example 6 *(Index 24: Gold: Sci/Tech, Predicted: Business)*

Rivals Try to Turn Tables on Charles Schwab By MICHAEL LIEDTKE SAN FRANCISCO (AP) – With its low prices and iconoclastic attitude, discount stock broker Charles Schwab Corp. (SCH) represented an annoying stone in Wall Street's wing-tipped shoes for decades…

Example 7 *(Index 36: Gold: World, Predicted: Business)*

Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market.

Example 8 *(Index 47: Gold: Sci/Tech, Predicted: Business)*

Dell Exits Low-End China Consumer PC Market HONG KONG (Reuters) - Dell Inc. ¡DELL.O¿, the world's largest PC maker, said on Monday it has left the low-end consumer PC market in China and cut its overall growth target for the country this year due to stiff competition in the segment.

Example 9 *(Index 55: Gold: Sci/Tech, Predicted: Sports)*

Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.

Example 10 *(Index 79: Gold: World, Predicted: Sports)*

Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.

Example 11 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

Example 12 *(Index 103: Gold: Business, Predicted: Sci/Tech)*

Surviving Biotech's Downturns Charly Travers offers advice on withstanding the volatility of the biotech sector.

Example 13 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space – which means that small businesses looking to move online get a sweeter deal through ...

Example 14 *(Index 120: Gold: World, Predicted: Business)*

Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 15 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 16 *(Index 183: Gold: Sci/Tech, Predicted: Business)*

Microsoft finalises three-year government deal Hot on the heels of its 10-year strategic partnership with the London Borough of Newham, Microsoft is close to signing a new broad three-year public sector agreement with the government.

Example 17 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.

Example 18 *(Index 209: Gold: Sci/Tech, Predicted: World)*

Alaska Wildfires Grow to Record 5 Million Acres (Reuters) Reuters - Wildfires have scorched over 5 million acres in Alaska as of Tuesday, forestry officials said, a new record that signals possible changes in climate conditions and the composition of the vast forests.

Example 19 *(Index 215: Gold: Sci/Tech, Predicted: Business)*

Hurricane center 's projection on Charley not far off, data show FORT LAUDERDALE, Fla. - (KRT) - Despite criticism that it should have better anticipated Hurricane Charley 's rapid intensification and quick turn, the National Hurricane Center 's forecast wasn 't that far off, a preliminary post-mortem shows.

Example 20 *(Index 222: Gold: Business, Predicted: Sci/Tech)*

Selling Houston Warts and All, Especially Warts Descriptions of urban afflictions and images of giant mosquitoes and cockroaches to convey a sense of how Houston is nevertheless beloved by many residents.

A.3 CNN (Max Sequence Length: 256)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.

Example 2 *(Index 8: Gold: Sci/Tech, Predicted: World)*

E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.

Example 3 *(Index 13: Gold: Sci/Tech, Predicted: World)*

Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.

Example 4 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.

Example 5 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of public participation. But this isn't the case.

Example 6 *(Index 24: Gold: Sci/Tech, Predicted: Business)*

Rivals Try to Turn Tables on Charles Schwab By MICHAEL LIEDTKE SAN FRANCISCO (AP) – With its low prices and iconoclastic attitude, discount stock broker Charles Schwab Corp. (SCH) represented an annoying stone in Wall Street's wing-tipped shoes for decades...

Example 7 *(Index 36: Gold: World, Predicted: Business)*

Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market.

Example 8 *(Index 47: Gold: Sci/Tech, Predicted: Business)*

Dell Exits Low-End China Consumer PC Market HONG KONG (Reuters) - Dell Inc. ¡DELL.O¿, the world's largest PC maker, said on Monday it has left the low-end consumer PC market in China and cut its overall growth target for the country this year due to stiff competition in the segment.

Example 9 *(Index 55: Gold: Sci/Tech, Predicted: Sports)*

Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.

Example 10 *(Index 79: Gold: World, Predicted: Sports)*

Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.

Example 11 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

Example 12 *(Index 103: Gold: Business, Predicted: Sci/Tech)*

Surviving Biotech's Downturns Charly Travers offers advice on withstanding the volatility of the biotech sector.

Example 13 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

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Example 14 *(Index 120: Gold: World, Predicted: Business)*

Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 15 *(Index 154: Gold: World, Predicted: Business)*

Google Lowers Its IPO Price Range SAN JOSE, Calif. - In a sign that Google Inc.'s initial public offering isn't as popular as expected, the company lowered its estimated price range to between $85 and $95 per share, down from the earlier prediction of $108 and $135 per share...

Example 16 *(Index 155: Gold: World, Predicted: Sci/Tech)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 17 *(Index 178: Gold: Sci/Tech, Predicted: Business)*

Oracle launches Business Integlligence 10g Oracle introduced a new BI platform yesterday, Business Intelligence 10g that rolls up into one solution all of their BI tools. However, more interesting than the nitty-gritty details of what is included is the back story taking place at the same time.

Example 18 *(Index 183: Gold: Sci/Tech, Predicted: Business)*

Microsoft finalises three-year government deal Hot on the heels of its 10-year strategic partnership with the London Borough of Newham, Microsoft is close to signing a new broad three-year public sector agreement with the government.

Example 19 *(Index 188: Gold: Sports, Predicted: Sci/Tech)*

Veterans Committee counts for little The Hall of Fame released the latest Veterans Committee ballot yesterday. As you might (or might not) remember, there 's a (nearly) new committee in town.

Example 20 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.

A.4 LSTM (Max Sequence Length: 64)

Total misclassified examples: 20

Example 1 *(Index 8: Gold: Sci/Tech, Predicted: World)*

E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.

Example 2 *(Index 9: Gold: Sci/Tech, Predicted: Business)*

Card fraud unit nets 36,000 cards In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests - and estimates it saved 65m.

Example 3 *(Index 19: Gold: Sci/Tech, Predicted: Business)*

Storage, servers bruise HP earnings update Earnings per share rise compared with a year ago, but company misses analysts' expectations by a long shot.

Example 4 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.

Example 5 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of public participation. But this isn't the case.

Example 6 *(Index 33: Gold: World, Predicted: Sports)*

Man Sought $50M From McGreevey, Aides Say (AP) AP - The man who claims Gov. James E. McGreevey sexually harassed him was pushing for a cash settlement of up to $50 million before the governor decided to announce that he was gay and had an extramarital affair, sources told The Associated Press.

Example 7 *(Index 47: Gold: Sci/Tech, Predicted: Business)*

Dell Exits Low-End China Consumer PC Market HONG KONG (Reuters) - Dell Inc. ¡DELL.O¿, the world's largest PC maker, said on Monday it has left the low-end consumer PC market in China and cut its overall growth target for the country this year due to stiff competition in the segment.

Example 8 *(Index 55: Gold: Sci/Tech, Predicted: Business)*

Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.

Example 9 *(Index 73: Gold: World, Predicted: Sci/Tech)*

On front line of AIDS in Russia An industrial city northwest of Moscow struggles as AIDS hits a broader population.

Example 10 *(Index 79: Gold: World, Predicted: Sports)*

Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.

Example 11 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Intel to delay product aimed for high-definition TVs SAN FRANCISCO – In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

Example 12 *(Index 84: Gold: World, Predicted: Business)*

Venezuela vote keeps Chavez as president CARACAS – Venezuelans voted resoundingly to keep firebrand populist Hugo Chavez as their president in a victory that drew noisy reactions yesterday from both sides in the streets. International observers certified the results as clean and accurate.

Example 13 *(Index 103: Gold: Business, Predicted: Sci/Tech)*

Surviving Biotech's Downturns Charly Travers offers advice on withstanding the volatility of the biotech sector.

Example 14 *(Index 104: Gold: World, Predicted: Sports)*

Mr Downer shoots his mouth off Just what Alexander Downer was thinking when he declared on radio last Friday that quot;they could fire a missile from North Korea to Sydney quot; is unclear. The provocative remark, just days before his arrival yesterday on his second visit to the North Korean ...

Example 15 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space – which means that small businesses looking to move online get a sweeter deal through ...

Example 16 *(Index 120: Gold: World, Predicted: Business)*

Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 17 *(Index 126: Gold: Sports, Predicted: World)*

Security scare as intruder dives in A CANADIAN husband 's love for his wife has led to a tightening of security at all Olympic venues in Athens.

Example 18 *(Index 136: Gold: World, Predicted: Sports)*

Schrder adopts Russian orphan Three-year-old Victoria, from St Petersburg, has been living at the Schrders ' family home in Hanover in northern Germany for several weeks.

Example 19 *(Index 154: Gold: World, Predicted: Business)*

Google Lowers Its IPO Price Range SAN JOSE, Calif. - In a sign that Google Inc.'s initial public offering isn't as popular as expected, the company lowered its estimated price range to between $85 and $95 per share, down from the earlier prediction of $108 and $135 per share...

Example 20 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

A.5 LSTM (Max Sequence Length: 128)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.

Example 2 *(Index 9: Gold: Sci/Tech, Predicted: Business)*

Card fraud unit nets 36,000 cards In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests - and estimates it saved 65m.

Example 3 *(Index 12: Gold: Sci/Tech, Predicted: Business)*

Dutch Retailer Beats Apple to Local Download Market AMSTERDAM (Reuters) - Free Record Shop, a Dutch music retail chain, beat Apple Computer Inc. to market on Tuesday with the launch of a new download service in Europe's latest battleground for digital song services.

Example 4 *(Index 15: Gold: Sci/Tech, Predicted: Business)*

Teenage T. rex's monster growth Tyrannosaurus rex achieved its massive size due to an enormous growth spurt during its adolescent years.

Example 5 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.

Example 6 *(Index 36: Gold: World, Predicted: Business)*

Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market.

Example 7 *(Index 41: Gold: Business, Predicted: World)*

Retailers Vie for Back-To-School Buyers (Reuters) Reuters - Apparel retailers are hoping their back-to-school fashions will make the grade among style-conscious teens and young adults this fall, but it could be a tough sell, with students and parents keeping a tighter hold on their wallets.

Example 8 *(Index 55: Gold: Sci/Tech, Predicted: Business)*

Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.

Example 9 *(Index 65: Gold: Sci/Tech, Predicted: Business)*

Indian state rolls out wireless broadband Government in South Indian state of Kerala sets up wireless kiosks as part of initiative to bridge digital divide.

Example 10 *(Index 79: Gold: World, Predicted: Sports)*

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Example 12 *(Index 104: Gold: World, Predicted: Sports)*

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Example 14 *(Index 111: Gold: Sci/Tech, Predicted: Business)*

IBM Buys Two Danish Services Firms IBM said Tuesday it has acquired a pair of Danish IT services firms as part of its effort to broaden its presence in Scandinavia. As a result of the moves, IBM will add about 3,700 IT staffers to its global head count. Financial terms of ...

Example 15 *(Index 120: Gold: World, Predicted: Business)*

Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 16 *(Index 125: Gold: Business, Predicted: World)*

SUVs: Live And Let Die NEW YORK - The newly released traffic crash fatality data have something for everyone in the debate about the safety of sport utility vehicles.

Example 17 *(Index 135: Gold: Sports, Predicted: World)*

US Men Have Right Touch in Relay Duel Against Australia THENS, Aug. 17 - So Michael Phelps is not going to match the seven gold medals won by Mark Spitz. And it is too early to tell if he will match Aleksandr Dityatin, the Soviet gymnast who won eight total medals in 1980. But those were not the ...

Example 18 *(Index 139: Gold: World, Predicted: Business)*

Sharon orders 1,000 homes in West Bank Israel announced plans for 1,000 houses in the West Bank yesterday, accelerating the expansion of the settlements.

Example 19 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 20 *(Index 172: Gold: Business, Predicted: Sci/Tech)*

Ross Stores Profit Plummets 40 Percent (AP) AP - Discount retailer Ross Stores Inc. Wednesday said its profit fell about 40 percent in the latest quarter due to problems with a new computer system that limited the company's ability to respond to changes in customer demand.

A.6 LSTM (Max Sequence Length: 256)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will roar.

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Example 8 *(Index 28: Gold: Sports, Predicted: World)*

Shoppach doesn't appear ready to hit the next level With the weeks dwindling until Jason Varitek enters free agency, the Red Sox continue to carefully monitor Kelly Shoppach , their catcher of the future, in his climb toward the majors. The Sox like most of what they have seen at Triple A Pawtucket from Shoppach, though it remains highly uncertain whether he can make the adjustments at the plate ...

Example 9 *(Index 36: Gold: World, Predicted: Business)*

Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil market.

Example 10 *(Index 55: Gold: Sci/Tech, Predicted: World)*

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Example 11 *(Index 56: Gold: World, Predicted: Business)*

India's Tata expands regional footprint via NatSteel buyout (AFP) AFP - India's Tata Iron and Steel Company Ltd. took a strategic step to expand its Asian footprint with the announcement it will buy the Asia-Pacific steel operations of Singapore's NatSteel Ltd.

Example 12 *(Index 65: Gold: Sci/Tech, Predicted: Business)*

Indian state rolls out wireless broadband Government in South Indian state of Kerala sets up wireless kiosks as part of initiative to bridge digital divide.

Example 13 *(Index 71: Gold: Sci/Tech, Predicted: Business)*

AOL Properties Sign Girafa For Thumbnail Search Images AOL Properties Sign Girafa For Thumbnail Search Images Girafa.com Inc. announced today that the CompuServe, Netscape, AIM and ICQ properties of America Online, Inc., have signed an agreement with Girafa to use Girafa's thumbnail search images as an integrated part of their search results. Using Girafa's thumbnail search service, search users can ...

Example 14 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

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Example 15 *(Index 89: Gold: Sci/Tech, Predicted: Business)*

Consumers Would Pay In Phone Proposal A proposal backed by a coalition of telephone carriers would cut billions of dollars in fees owed by long-distance companies to regional phone giants but would allow the regional companies to make up some of the difference by raising monthly phone bills for millions of consumers. ¡FONT face="verdana,MS Sans Serif,arial,helvetica" size="-2" color="#666666"¿¡B¿-The Washington Post¡/B¿¡/FONT¿

Example 16 *(Index 94: Gold: Sports, Predicted: World)*

Kederis proclaims innocence Olympic champion Kostas Kederis today left hospital ahead of his date with IOC inquisitors claiming his innocence and vowing: quot;After the crucifixion comes the resurrection. quot; ...

Example 17 *(Index 103: Gold: Business, Predicted: Sci/Tech)*

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Example 18 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

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